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Flow of Control

Learning Objectives

After learning this chapter, the students will be able to

- Understand the different kinds of statements.
- Construct different flow of control statements in C++.

10.1 Introduction

In the previous chapters you learnt the basic concepts of C++ programming such as variables, constants, operators, data types etc. Generally a program executes its statements sequentially from beginning to end. However, such a strict sequential ordering is restrictive and less useful. There are lot of situations where it is useful to decide the code block executed on the basis of a certain condition. In such situations, the flow of control jumps from one part of the code to another segment of code. Program statements that cause such jumps are called as **"Control flow"**. This chapter deals with the basics of control structures such as "Selection", "Iteration" and "Jump" statement.

10.2 Statements

A computer program is a set of statements or instructions to perform a specific task. These statements are intended to perform specific action. The action may be of variable declarations, expression evaluations, assignment operations, decision making, looping and so on.

There are two kinds of statements used in C++.

(i) Null statement

(ii) Compound statement

10.2.1 Null statement

The "**null** or **empty statement**" is a statement containing only a semicolon. It takes the flowing form:

```
; // it is a null statement
```

Null statements are commonly used as placeholders in iteration statements or as statements on which to place labels at the end of compound statements or functions.

10.2.2 Compound (Block) statement

C++ allows a group of statements enclosed by pair of braces {}. This group of statements is called as a compound statement or a block.

The general format of compound statement is: